Statistical Tests v/s Exploratory Data Analysis

Understanding the differences between statistical tests and exploratory data analysis (EDA) is crucial for effective data analysis and decision-making.

Exploratory Data Analysis (EDA)

01 **Objective**: Gain insights on data characteristics and relationships

02 **Purpose**: Explore and understand data without assumptions

03 **Methods**: Statistical graphics, plots, info tables, summary stats

04 **Nature**: Descriptive, visual, interactive

Statistical Tests

01 **Objective**: Provide quantitative evidence for hypotheses

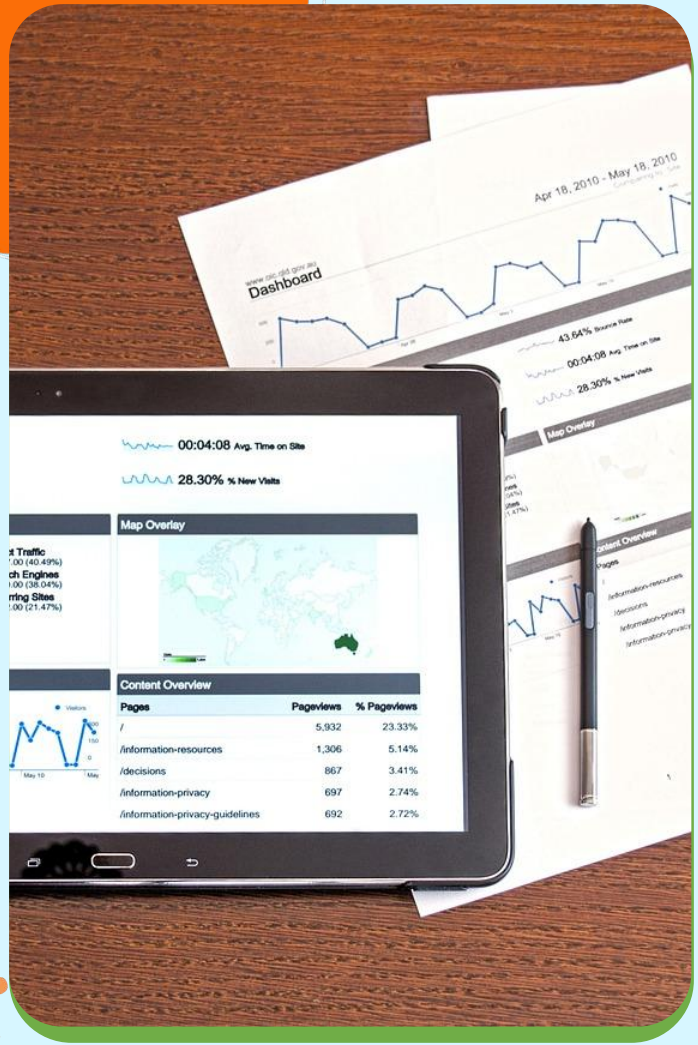
02 **Purpose**: Make inferences about populations based on sample data

03 **Methods**: Mathematical formulas, models, hypothesis testing

04 **Nature**: Formal, hypothesis-driven

Summary

- EDA: Understand and summarize data
- Statistical tests: Test hypotheses and make inferences
- EDA precedes statistical testing
- EDA guides hypothesis formulation



Thanks for watching. 😄